RV College of Engineering®

Go, change the world

Unit 3(4NF and 5NF)

Original Content:
Ramez Elmasri and Shamkant B. Navathe

Dr. Poonam Ghuli
Associate Professor, Department of CSE
RV College of Engineering, Bengaluru - 59

Fourth Normal Form

- Fourth Normal Form comes into picture when **Multi-valued Dependency** occur in any relation.
- For a Relation to satisfy the Fourth Normal Form, it should satisfy the following two conditions:
    - It should be in the **Boyce-Codd Normal Form**.
    - And, the table should not have any **Multi-valued Dependency**.

What is Multi-valued Dependency?

- A Relation is said to have multi-valued dependency, if the following 3 conditions are true,
  - For a dependency A → B, if **for a single value of A, multiple value of B exists**, then the table may have multi-valued dependency.
  - Also, a table should have **at-least 3 columns** for it to have a multi-valued dependency.
  - And, for a relation R(A,B,C), if there is a multi-valued dependency between A and B, then **B and C attributes should be independent of each other**.
- **MVD between attributes A, B, and C in a relation using the following notation:**

A –>> B

A –>> C

Trivial MVD

- **A multi-valued dependency can be further defined as being trivial or nontrivial.**
  - **A MVD A –>> B in relation R is defined as being trivial if**

    (a) B is a subset of A *or*

    (b) A ∪ B = R.
  - **A MVD is defined as being nontrivial if neither (a) nor (b) are satisfied.**
  - **A trivial MVD does not specify a constraint on a relation, while a nontrivial MVD does specify a constraint.**

Fourth Normal Form - Definition

- **A relation is in 4NF if it is in Boyce-Codd Normal Form and contains no nontrivial multi-valued dependencies.**

Definition:

- A relation schema *R* is in **4NF** with respect to a set of dependencies *F* (that includes functional dependencies and multivalued dependencies) if, for every *nontrivial* multivalued dependency $X \twoheadrightarrow Y$ in F^+, *X* is a superkey for R.

Note: F^+ is the (complete) set of all dependencies (functional or multivalued) that will hold in every relation state *r* of *R* that satisfies *F*. It is also called the **closure** of *F*.

Fourth Normal Form - Example

- As you can see in the table above, student with s_id **1** has opted for two courses, **Science** and **Maths**, and has two hobbies, **Cricket** and **Hockey**.
- Well the two records for student with s_id **1**, will give rise to two more records, as shown below, because for one student, two hobbies exists, hence along with both the courses, these hobbies should be specified.

s_id	course	hobby
1	Science	Cricket
1	Maths	Hockey
2	C#	Cricket
2	Php	Hockey

Fourth Normal Form - Example

- And, in the table below, there is no relationship between the columns course and hobby. They are independent of each other.
- So there is multi-value dependency, which leads to un-necessary repetition of data and other anomalies as well.

s_id	course	hobby
1	Science	Cricket
1	Maths	Hockey
1	Science	Hockey
1	Maths	Cricket

How to satisfy 4th Normal Form?

- To make the above relation satisfy the 4th normal form, we can decompose the table into 2 tables.
- A table can also have functional dependency along with multi-valued dependency. In that case, the functionally dependent columns are moved in a separate table and the multi-valued dependent columns are moved to separate tables.

CourseOpted Table

s_id	course
1	Science
1	Maths
2	C#
2	Php

And, **Hobbies Table**,

s_id	hobby
1	Cricket
1	Hockey
2	Cricket
2	Hockey

Fifth Normal Form (5NF)

- Previous slides have pointed out the problematic functional dependencies and shown how they were eliminated by a process of repeated binary decomposition during the process of normalization to achieve 1NF, 2NF, 3NF, and BCNF.

- Achieving 4NF typically involves eliminating MVDs by repeated binary decompositions as well.

Fifth Normal Form (5NF)

- However, in some cases there may be no nonadditive join decomposition of *R* into *two relation schemas, but there may be a nonadditive join decomposition into more than two relation schemas.*

- This type of another dependency called the *join dependency and, if it is present, carry out a multiway decomposition* into fifth normal form (5NF).

- It is important to note that such a dependency is a peculiar semantic constraint that is difficult to detect in practice; therefore, normalization into 5NF is rarely done in practice.

Fourth Normal Form (4NF) with Functional Dependency Example (Repeated Binary Decomposition)

R(SID, Course, Email, DOB, Address, Name, Hobby)

R1 (SID,Course)

R2(SID,Email,DOB,Address,Name,Hobby)

R21(SID,Hobby)

R22(SID,Email,DOB,Address,Name)

Fifth Normal Form (5NF)

Definition:

- A **join dependency** (**JD**), denoted by JD(R_1, R_2, ..., R_n), specified on relation schema R, specifies a constraint on the states r of R. The constraint states that every legal state r of R should have a non-additive join decomposition into R_1, R_2, ..., R_n; that is, for every such r we have

$$* (\pi_{R1}(r), \pi_{R2}(r), ..., \pi_{Rn}(r)) = r$$

- A join dependency JD(R_1, R_2, ..., R_n), specified on relation schema R, is a **trivial JD** if one of the relation schemas R_i in JD(R_1, R_2, ..., R_n) is equal to R.

Fifth Normal Form (5NF)

Definition:

- A relation schema *R* is in **fifth normal form** (**5NF**) (or **Project-Join Normal Form** (**PJNF**)) with respect to a set *F* of functional, multivalued, and join dependencies if,
  - R should be already in 4NF.
  - It cannot be further non loss decomposed (join dependency)

- Discovering JDs in practical databases with hundreds of attributes is next to impossible.
- It can be done only with a great degree of intuition about the data on the part of the designer. Therefore, the current practice of database design pays scant attention to them.

Fifth Normal Form (5NF) - Example

- The central concept of 5NF can be explained with a commonly used example, namely involving attributes agents, companies, and products.
- If agents represent companies, companies make products, and agents sell products, then we might want to keep a record of which agent sells which product for which company. This information could be kept in one record type with three fields:

Agent	Company	Product
Smith	Ford	Car
Smith	Ford	Trucks
Harry	Ford	Bus
Smith	GM	Bus

Fifth Normal Form (5NF)

- This form is necessary in the general case. For example, although agent Smith sells cars and trucks made by Ford and Buses made by GM, he does not sell Ford Buses.

- The table is in 4NF because it contains no multi-valued dependency. It does, however, contain an element of redundancy in that it records the fact that Smith is an agent for Ford twice.

- But there is no way of eliminating this redundancy without losing information. Suppose that the table is decomposed into its two projections, PI and P2.
  - P1(Agent, Company) and P2(Company, Product)
  - P1*P2 will result in spurious tuple generation

Fifth Normal Form (5NF)

- Now suppose that the original table were to be decomposed into three tables, the two projections, P I and P2 which have already shown, and the final, possible projection, P3.
  - P1(Agent, Company) , P2(Company, Product) and P3(Agent, Product)
  - Still the natural join of all three projections P1*P2*P3 will result in spurious tuple generation

- The order in which the joins are performed makes no difference to the final result. It is not simply possible of decompose the 'AGENT_COMPANY_PRODUCT' table, populated as shown, without losing information.
- **Thus, you have to retain the original table. You cannot decompose further**

Fifth Normal Form (5NF)

- Thus, we need the combination of all three fields to know which combinations are valid and which are not.
- But suppose that a certain rule was in effect: **if an agent sells a certain product, and if he represents a company making that product, then he sells that product for that company.**

Agent	Company	Product
Smith	Ford	Car
Smith	Ford	Trucks
Harry	Ford	Bus
Smith	GM	Bus
Smith	Ford	Bus

Fifth Normal Form (5NF)

- In this case, it turns out that we can reconstruct all the true facts from a normalized form consisting of three separate record types (Projections P1, P2 and P3), each containing two fields:
  - P1(Agent, Company) , P2(Company, Product) and P3(Agent, Company)
  - Natural join of all three projections P1*P2*P3 will result in generation of original facts

- Fifth normal form does not differ from fourth normal form unless there exists a symmetric constraint such as the rule about agents, companies, and products.

- In the absence of such a constraint, a record type in fourth normal form is always in fifth normal form.

Fifth Normal Form (5NF)

- It should be observed that although the normalized form involves more record types, there may be fewer total record occurrences.

- This is not apparent when there are only a few facts to record, as in the example shown above. The advantage is realized as more facts are recorded, since the size of the normalized files increases in an additive fashion, while the size of the unnormalized file increases in a multiplicative fashion.

- For example, if we add a new agent who sells x products for y companies, where each of these companies makes each of these products, we have to add x+y new records to the normalized form, but xy new records to the unnormalized form.

Fifth Normal Form (5NF)

- It is important to note that such a dependency is a peculiar semantic constraint that is difficult to detect in practice; therefore, normalization into 5NF is rarely done in practice.

- Fifth normal form is needed in order to deal with the redundancies and eliminate them to a certain extent.

Thank YOU

Check for lossless join Decomposition

https://drive.google.com/file/d/1HfVJoTjsVswO9adBdFnNYatI3kGDuV0o/view?usp=sharing

4NF and 5NF

https://drive.google.com/file/d/1Dq-k_72hTTkfbCsajomwpdw6AQ58Vi5E/view?usp=sharing